Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: AM3 Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
836 W Linwood AVE	Crows Landing	Stanislaus	95313

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X005-X011-XXXX X058-X022-X012-XXXX

B. OPERATORS

Mendes, Albert

Operator name: Mendes, Albert Telephone no.: (209) 678-0272 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
1100 Ruble RD	Crows Landing	CA	95313

This operator is responsible for paying permit fees.

C. OWNERS

Mendes, Albert

Legal owner name: Mendes, Albert Telephone no.: (209) 678-0272 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
1100 Ruble RD	Crows Landing	CA	95313

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	100	300	0	0	0
Number under roof	575	0	0	0	0	0
Maximum number	575	100	300	0	0	0
Average number	575	100	300	0	0	0
Avg live weight (lbs)	1,400	1,400	900	0		

Predominant milk cow breed: Holstein

Average milk production: 75 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 19,567.58 *tons per reporting period*

Total nitrogen from manure: 243,975.13 *lbs per reporting period*

After ammonia losses (30% loss applied): 170,782.59 *lbs per reporting period*

Total phosphorus from manure: 40,685.42 *lbs per reporting period*

Total potassium from manure: 112,006.68 *lbs per reporting period*

Total salt from manure: 293,733.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,053,093 *gallons*

Total nitrogen generated: 31,996.40 *lbs*

Total phosphorus generated: 2,624.15 *lbs*

Total potassium generated: 28,570.17 *lbs*

Total salt generated: 500,311.74 *lbs*

	7,053,093	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	7,053,093	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
East Irrigation Well	Ground water
TID Lateral 4	Ground water
West Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/30/2023	Solid commercial fertilizer *NH3*	4.80 *ton*	Dry-weight	0.1	82.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	7,864.13	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	7,864.13	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	Separator solids	1,032.00 *ton*	Dry-weight	59.6		17,200.00	7,320.00	45,000.00		42.10

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	14,342.32	6,103.83	37,523.52	351,053.38
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	14,342.32	6,103.83	37,523.52	351,053.38

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
D22	22	22	2	process wastewater	X058-X022-X011-XXXX
H40	40	40	2	process wastewater	X058-X022-0004-XXXX
S18	18	18	2	process wastewater	X058-X022-X034-XXXX
Totals for areas that were used for application	80	80	6		
Totals for areas that were not used for application					
Land application area totals	80	80	6		

B. CROPS AND HARVESTS

D22

Field name: D22

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/06/2023	451.00 *ton*	As-is		70.0	3,990.00	1,040.00	11,100.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	20.50	163.59	42.64	455.10	1,340.70

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	672.00 *ton*	As-is		70.0	4,410.00	560.00	4,870.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	30.55	269.41	34.21	297.51	1,191.27

H40

Field name: H40

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	757.00 *ton*	As-is		70.0	4,500.00	1,020.00	8,280.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	18.93	170.33	38.61	313.40	1,180.92

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,029.00 *ton*	As-is		70.0	4,200.00	680.00	5,160.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.73	216.09	34.99	265.48	1,049.58

S18

Field name: S18

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	412.00 *ton*	As-is		70.0	4,020.00	880.00	9,000.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	22.89	184.03	40.28	412.00	1,442.00

S18

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	545.00 *ton*	As-is		70.0	3,960.00	600.00	4,800.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	30.28	239.80	36.33	290.67	1,199.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

D22 - 11/05/2022: Oats, silage-soft dough

Field name: D22

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.11	6.01	68.60	938.64	201,182.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	1,782,000.00 *gal*
Application event totals		44.11	6.01	68.60	956.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.99	7.63	87.09	1,191.53	255,385.00 *gal*
East Irrigation Well	Ground water	3.56	0.00	0.00	115.93	375,000.00 *gal*
Application event totals		59.55	7.63	87.09	1,307.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irrigation Well	Ground water	0.96	0.00	0.00	318.63	600,000.00 *gal*
Application event totals		0.96	0.00	0.00	318.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.76	3.40	32.97	720.39	201,182.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	1,782,000.00 *gal*
Application event totals		49.76	3.40	32.97	737.96	

D22 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.16	4.31	41.85	914.47	255,385.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	1,782,000.00 *gal*
Application event totals		63.16	4.31	41.85	932.05	

D22 - 05/31/2023: Corn, silage

Field name: D22

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.39	8.49	82.42	1,800.96	502,955.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	47.93	4,860,000.00 *gal*
Application event totals		124.39	8.49	82.42	1,848.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
TID Lateral 4	Ground water	0.00	0.00	0.00	23.97	2,430,000.00 *gal*
Application event totals		49.20	0.00	0.00	23.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.97	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.97	

D22 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	27.96	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.96	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.85	0.99	22.70	408.27	502,955.00 *gal*
Application event totals		52.85	0.99	22.70	408.27	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.42	0.50	11.35	204.13	251,478.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	19.97	2,025,000.00 *gal*
Application event totals		26.42	0.50	11.35	224.11	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	19.97	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.97	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	15.98	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.98	

H40 - 11/06/2022: Oats, silage-soft dough

Field name: H40

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.65	8.27	94.33	1,290.63	502,955.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	27.39	5,049,000.00 *gal*
Application event totals		60.65	8.27	94.33	1,318.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.65	8.27	94.33	1,290.63	502,955.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	16.11	2,970,000.00 *gal*
Application event totals		60.65	8.27	94.33	1,306.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
East Irrigation Well	Ground water	3.13	0.00	0.00	102.02	600,000.00 *gal*
West Irrigation Well	Ground water	0.84	0.00	0.00	280.39	960,000.00 *gal*
Application event totals		3.97	0.00	0.00	382.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.41	4.67	45.33	990.53	502,955.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	19.33	3,564,000.00 *gal*
Application event totals		68.41	4.67	45.33	1,009.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.37	1.87	18.13	396.21	201,182.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	19.33	3,564,000.00 *gal*
Application event totals		27.37	1.87	18.13	415.54	

H40 - 06/06/2023: Corn, silage

Field name: H40

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	136.83	9.34	90.66	1,981.06	1,005,910.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	35.44	6,534,000.00 *gal*
Application event totals		136.83	9.34	90.66	2,016.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	19.33	3,564,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	22.55	4,158,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

H40 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	22.55	4,158,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	19.33	3,564,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.07	0.55	12.49	224.55	502,955.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	22.55	4,158,000.00 *gal*
Application event totals		29.07	0.55	12.49	247.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	22.55	4,158,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.55	

S18 - 11/06/2022: Oats, silage-soft dough

Field name: S18

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

S18 - 11/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.43	9.33	106.44	1,456.31	255,385.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.48	1,782,000.00 *gal*
Application event totals		68.43	9.33	106.44	1,477.79	

01/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.91	7.35	83.85	1,147.22	201,182.00 *gal*
East Irrigation Well	Ground water	4.35	0.00	0.00	141.69	375,000.00 *gal*
Application event totals		58.26	7.35	83.85	1,288.91	

01/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irrigation Well	Ground water	1.17	0.00	0.00	389.43	600,000.00 *gal*
Application event totals		1.17	0.00	0.00	389.43	

02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.81	4.15	40.29	880.47	201,182.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.90	1,485,000.00 *gal*
Application event totals		60.81	4.15	40.29	898.37	

03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.81	4.15	40.29	880.47	201,182.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.90	1,485,000.00 *gal*
Application event totals		60.81	4.15	40.29	898.37	

S18 - 11/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	14.32	1,188,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.32	

S18 - 05/31/2023: Corn, silage

Field name: S18

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.62	8.30	80.59	1,760.94	402,364.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	58.58	4,860,000.00 *gal*
Application event totals		121.62	8.30	80.59	1,819.52	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

06/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

S18 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.84	0.49	11.10	199.60	201,182.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	2,430,000.00 *gal*
Application event totals		25.84	0.49	11.10	228.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.84	0.49	11.10	199.60	201,182.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	24.41	2,025,000.00 *gal*
Application event totals		25.84	0.49	11.10	224.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

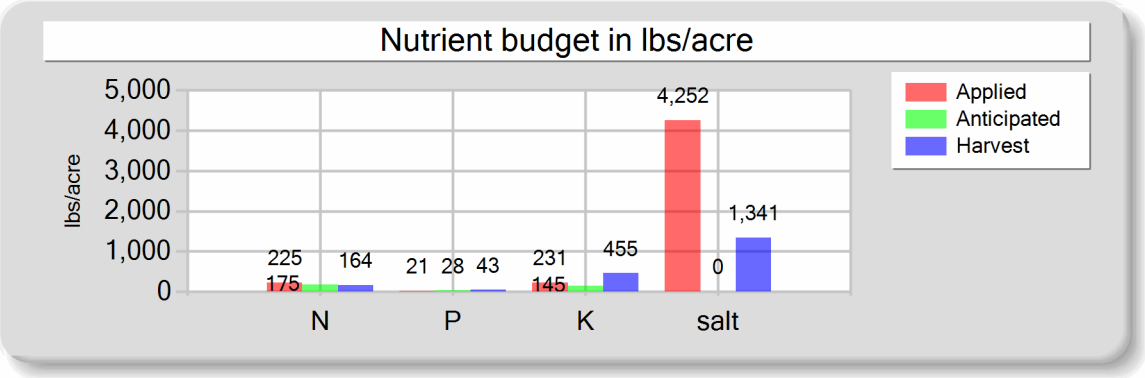
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	19.53	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.53	

B. NUTRIENT BUDGET

D22 - 11/05/2022: Oats, silage-soft dough

Field name: D22 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.02	21.35	230.51	3,765.02
Fresh water	4.51	0.00	0.00	487.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.53	21.35	230.51	4,252.30
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	163.59	42.64	455.10	1,340.70
Nutrient balance	60.94	-21.29	-224.59	2,911.60
Applied to removed ratio	1.37	0.50	0.51	3.17

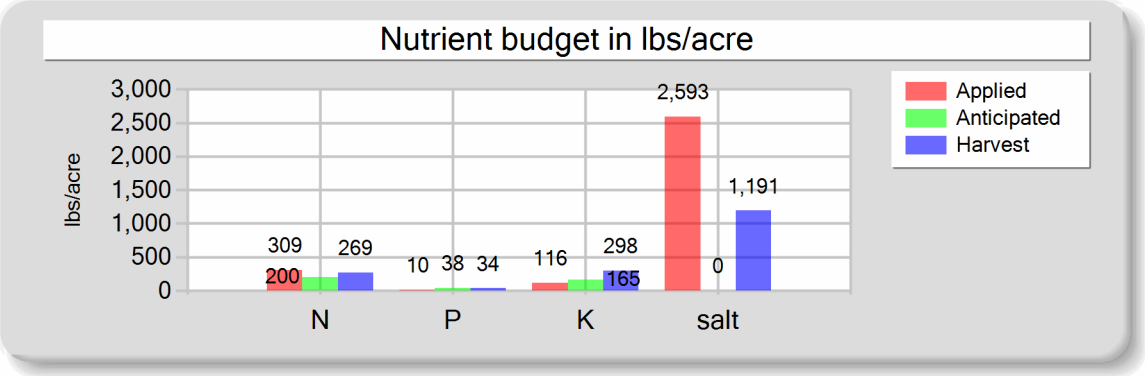
Fresh water applied	
6,321,000.00	*gallons*
232.78	*acre-inches*
10.58	*inches/acre*

Process wastewater applied	
913,134.00	*gallons*
33.63	*acre-inches*
1.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

D22 - 05/31/2023: Corn, silage

Field name: D22 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	98.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	203.66	9.98	116.47	2,413.37
Fresh water	0.00	0.00	0.00	179.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	309.06	9.98	116.47	2,593.11
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	269.41	34.21	297.51	1,191.27
Nutrient balance	39.65	-24.23	-181.04	1,401.83
Applied to removed ratio	1.15	0.29	0.39	2.18

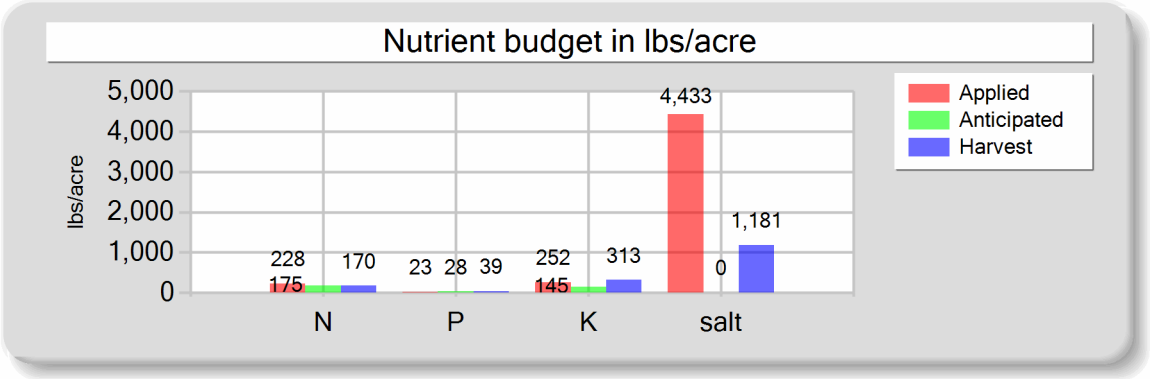
Fresh water applied	
18,225,000.00	*gallons*
671.16	*acre-inches*
30.51	*inches/acre*

Process wastewater applied	
1,257,388.00	*gallons*
46.31	*acre-inches*
2.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

H40 - 11/06/2022: Oats, silage-soft dough

Field name: H40 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.08	23.07	252.12	3,967.99
Fresh water	3.97	0.00	0.00	464.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.05	23.07	252.12	4,432.57
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	170.33	38.61	313.40	1,180.92
Nutrient balance	57.72	-15.53	-61.27	3,251.65
Applied to removed ratio	1.34	0.60	0.80	3.75

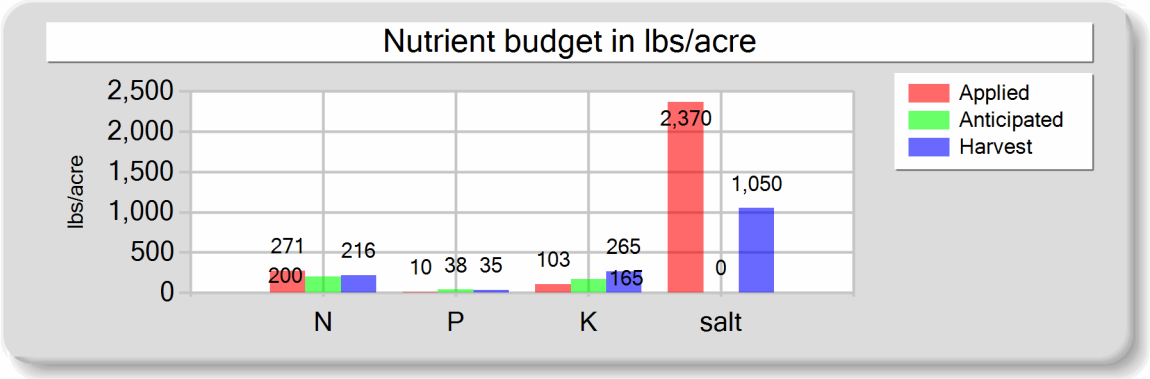
Fresh water applied	
16,707,000.00	*gallons*
615.26	*acre-inches*
15.38	*inches/acre*

Process wastewater applied	
1,710,047.00	*gallons*
62.98	*acre-inches*
1.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

H40 - 06/06/2023: Corn, silage

Field name: H40 Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	98.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	165.89	9.88	103.15	2,205.61
Fresh water	0.00	0.00	0.00	164.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.29	9.88	103.15	2,369.93
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	216.09	34.99	265.48	1,049.58
Nutrient balance	55.20	-25.10	-162.34	1,320.35
Applied to removed ratio	1.26	0.28	0.39	2.26

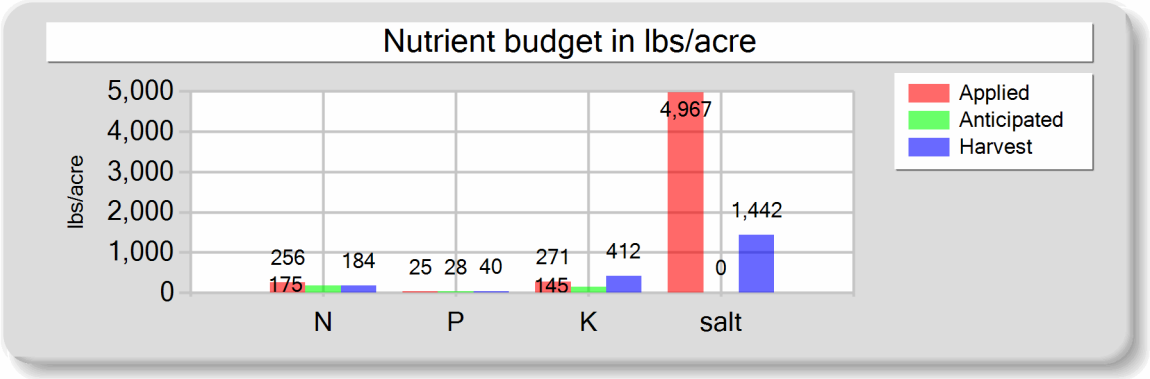
Fresh water applied	
30,294,000.00	*gallons*
1,115.63	*acre-inches*
27.89	*inches/acre*

Process wastewater applied	
1,508,865.00	*gallons*
55.57	*acre-inches*
1.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

S18 - 11/06/2022: Oats, silage-soft dough

Field name: S18 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	243.97	24.98	270.88	4,364.48
Fresh water	5.52	0.00	0.00	602.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	256.49	24.98	270.88	4,967.20
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	184.03	40.28	412.00	1,442.00
Nutrient balance	72.46	-15.30	-141.12	3,525.20
Applied to removed ratio	1.39	0.62	0.66	3.44

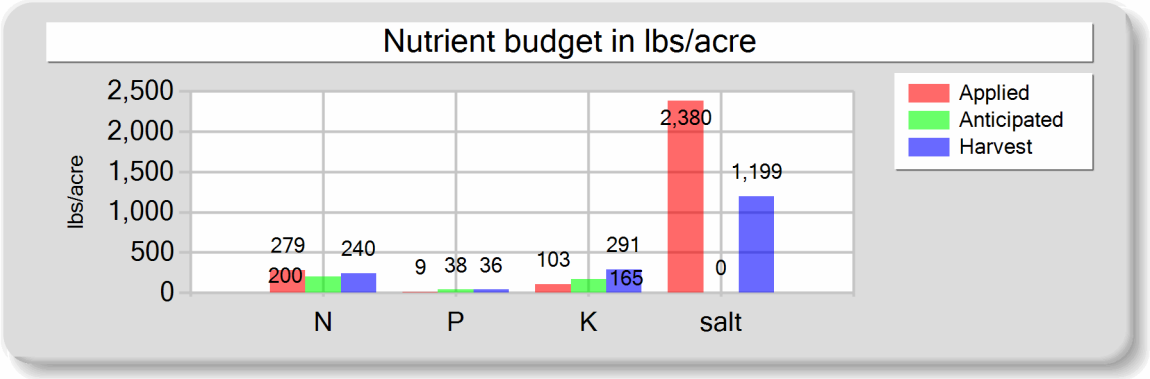
Fresh water applied	
6,915,000.00	*gallons*
254.66	*acre-inches*
14.15	*inches/acre*

Process wastewater applied	
858,931.00	*gallons*
31.63	*acre-inches*
1.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

S18 - 05/31/2023: Corn, silage

Field name: S18 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	98.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.30	9.27	102.78	2,160.14
Fresh water	0.00	0.00	0.00	219.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.70	9.27	102.78	2,379.82
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	239.80	36.33	290.67	1,199.00
Nutrient balance	38.90	-27.06	-187.88	1,180.82
Applied to removed ratio	1.16	0.26	0.35	1.98

Fresh water applied	
18,225,000.00	*gallons*
671.16	*acre-inches*
37.29	*inches/acre*

Process wastewater applied	
804,728.00	*gallons*
29.64	*acre-inches*
1.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

No manure analyses entered.

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/04/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	578.00	387.00	0.00	0.00	78.80	899.00								8,410.00	12,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,150.00	428.00	0.00	0.00	116.00	894.00								7,970.00	8,580
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWS

Sample and source description: WWS

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	652.00	378.00	0.00	0.00	44.50	432.00								8,750.00	9,440
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 07/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	277.00	58.80	0.00	0.00	5.20	119.00								3,290.00	2,140
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 10/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	246.00	154.00	0.00	0.00	35.10	272.00								5,000.00	3,620
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

East Irrigation Well

Fresh Water

Sample description: Fresh Water

Sample date: 12/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	25.00	0.00	25.00								2,000.00	815
DL	0.50	0.50	0.50								1.00	10

TID Lateral 4

TID Lateral 4

Fresh Water

Sample description: Fresh Water

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.00	26
DL	0.50	0.50	0.50								1.00	10

West Irrigation Well

Fresh Water

Sample description: Fresh Water

Sample date: 12/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.21	0.00	4.21								2,680.00	1,400
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

D22 - 11/05/2022: Oats, silage-soft dough

D22 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,990.00	1,040.00	11,100.00		10.90
DL	100.00	100.00	100.00		0.01

D22 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,410.00	560.00	4,870.00		6.50
DL	100.00	100.00	100.00		0.01

H40 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	1,020.00	8,280.00		10.40
DL	100.00	100.00	100.00		0.01

H40 - 06/06/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	680.00	5,160.00		6.80
DL	100.00	100.00	100.00		0.01

S18 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,020.00	880.00	9,000.00		10.50
DL	100.00	100.00	100.00		0.01

S18 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,960.00	600.00	4,800.00		6.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

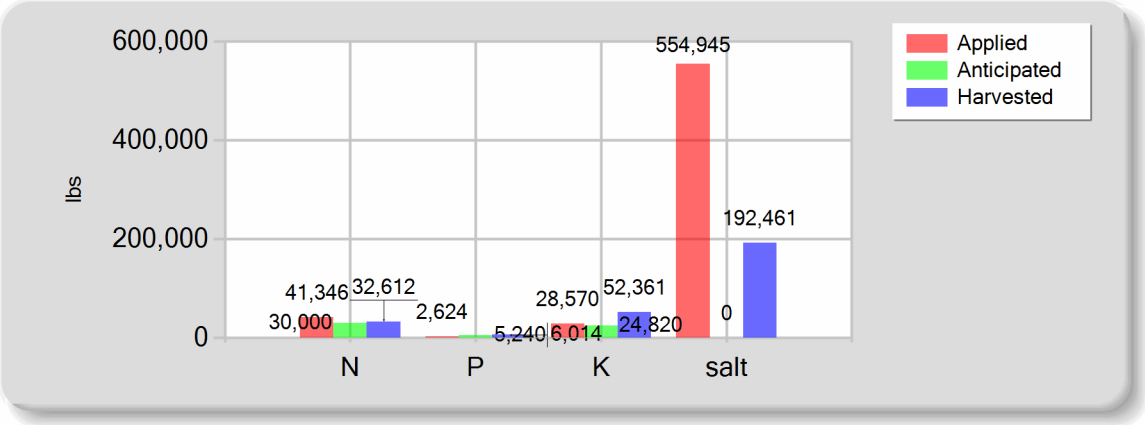
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

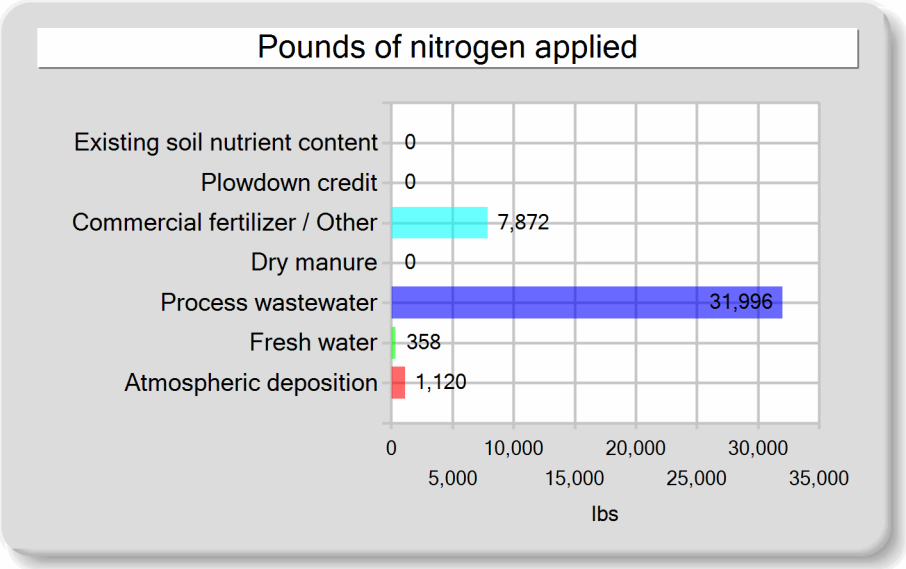
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

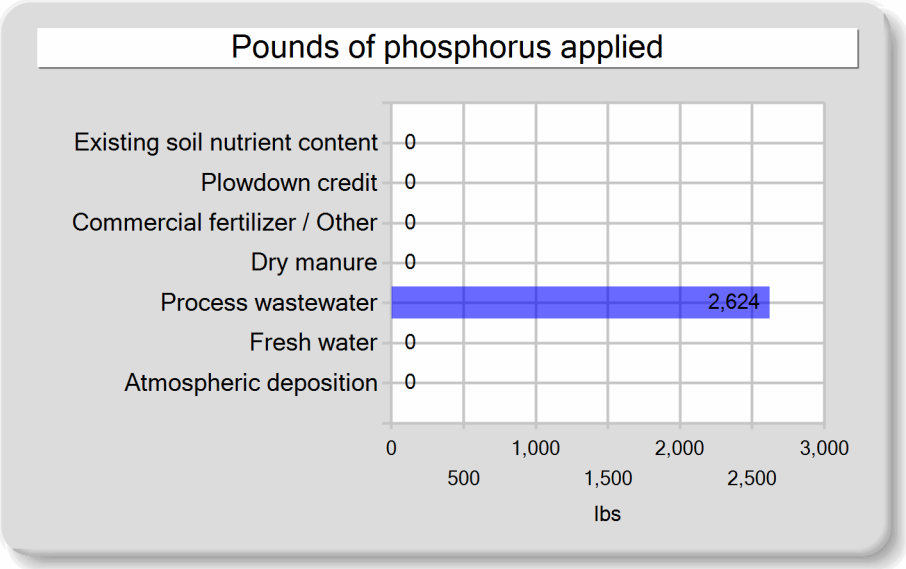
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	7,872.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	31,996.40	2,624.15	28,570.17	500,311.74
Fresh water	357.53	0.00	0.00	54,633.48
Atmospheric deposition	1,120.00	0.00	0.00	0.00
Total nutrients applied	41,345.93	2,624.15	28,570.17	554,945.22
Anticipated crop nutrient removal	30,000.00	5,240.00	24,820.00	0.00
Actual crop nutrient removal	32,611.50	6,013.56	52,360.68	192,461.40
Nutrient balance	8,734.43	-3,389.41	-23,790.51	362,483.82
Applied to removed ratio	1.27	0.44	0.55	2.88

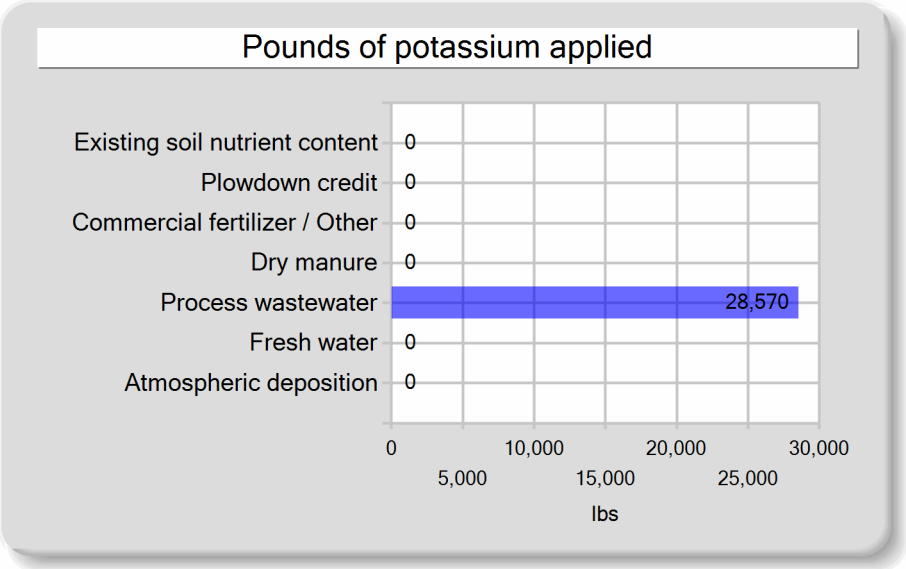
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

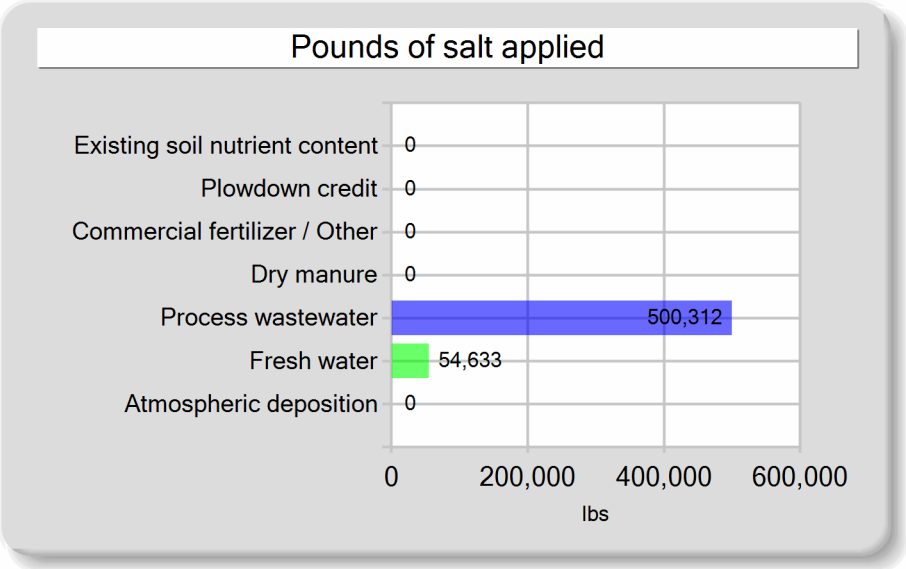


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Albert Mendes (Apr 19, 2024 04:42 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Albert Mendes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 19, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Albert Mendes

Name of Dairy Facility: AM3 Dairy

Facility Address:

Number and Street	City	County	Zip Code
836 W Linwood AVE	Crows Landing	Stanislaus	95313

Contact Person Name and Phone Number:

Name	Phone Number
Albert Mendes	(209) 678-0272

MANURE HAULER INFORMATION

Name of Hauling Company/Person: 5 Star Custom Spreading

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
219 Jensen Rd	Gustine	CA	95322

Contact Person:

Name	Phone Number
Eric Rocha	(209) 854-6251

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Ruble Ranch	(209) 678-0272

Address	City	State	Zip Code
1100 Ruble Rd	Crows Landing	CA	95313

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code

Street and nearest cross street (if no address)	County

Assessor's Parcel Number	Assessor's Parcel Number County
0058-0004-0009-0010	Merced

Last date hauled: 09/15/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,032.00 tons

Manure Solids Content: 40.4 %

Method used to determine amount of manure:

Haulers Calculations

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Albert Mendes (Apr 19, 2024 04:42 PDT)

Apr 19, 2024

Operator Signature | Date

Eric Rocha

Hauler Signature | Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Mendes Dairy
1100 Ruble Rd.
Crowslanding, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: AM3
RECEIVED DATE: 12/18/23 2:05 PM
SUBMITTED BY:
ANALYZED DATE: 12/22/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Small Well	T0356009A	Electrical Conductivity (EC)	EPA 120.1 and	3.15	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.64	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-2011	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2020	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	102 (null)	102 (null)	0.000
Electrical Conductivity (EC)	101			0.000
Nitrate Nitrogen (NO3-N)				

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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Compliance Analysis Report

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Mendes Dairy
1100 Ruble Rd.
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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: AM3
RECEIVED DATE: 12/18/23 2:05 PM
SUBMITTED BY:
ANALYZED DATE: 12/22/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Big Well	T0356009B	Electrical Conductivity (EC)	EPA 120.1 and	3.38	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.81	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-2011	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2160	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	102 (null)	102 (null)	0.000
Electrical Conductivity (EC)	101			0.000
Nitrate Nitrogen (NO3-N)				

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: AM3 Dairy
Mailing address: 1100 Buble Rd
(not facility address)
Phone#:

Authorized Copy Released To: F.B. Hy
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- ~~H3 EC*, NH4-N, TKN, TP, K, TDS~~
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12-17	10:30	Small Well		16.2°		X				
2		10:35	Big Well		16.4°		X				
3		10:40	lagoon		16.2°						K8
4		10:45	Solid manure								M7
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	[signature]					
2nd	[signature]	DENELE ANALYTICAL, INC.	12/18/23 2:05			T0356009

AMD 2023 Annual Report Complete

Final Audit Report 2024-04-19

Created:	2024-04-17
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAJd6TO86aG1WJHuU4P3-E0OOAnhzroezZ

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2024-04-19 - 11:42:55 AM GMT

Document e-signed by Albert Mendes (bertm55@msn.com)
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